

Product Requirement Document (PRD)

CampusFind Pro - Comprehensive Product Specifications, Personas, and Roadmap

1. Project Overview & Business Value

In academic settings, students frequently lose essential personal and study belongings, such as laptops, calculators, lab notebooks, backpacks, and wallets. The manual notification boards traditionally used by universities are inefficient, slow, insecure, and lack auditing capabilities. CampusFind Pro solves this operational gap by introducing a secure, centralized, and cloud-indexed Lost & Found portal. The business value lies in reducing property loss, decreasing load at administrative support desks, and providing students with a secure platform to claim and recover items.

2. Target Audience & Detailed Personas

The application targets three distinct campus stakeholder groups, each mapped to a concrete persona:

- **Student (Dev Darji):** devdarji@university.edu. Dev is a busy undergraduate student who needs a fast way to check if someone turned in his lost calculator. He wants one-click Google OAuth access, real-time alerts, and a dashboard where he can track his claims without having to constantly visit the campus security office.
- **Campus Security Officer (Admin):** officer.smith@university.edu. Officer Smith needs an interface to manage items turned in, verify claims of ownership (by validating descriptions, matching receipts, or checking security answers), and approve/reject claims securely.
- **IT Administrator (System Admin):** admin@campusfindpro.edu. This persona manages database seeds, configures API variables (such as Resend or Brevo credentials), reviews activity logs for audits, and ensures that the platform compliance matches FERPA guidelines.

3. Comprehensive Functional Requirements

- **Google OAuth Authentication:** One-click login that checks user domains, automatically provisions student roles, and syncs names and remote profile avatars on load.
- **Lost/Found Items Submission:** Form validating titles, descriptions, categories, exact campus location markers, lost/found dates, reward structures, and file uploads.
- **Verification and Claim Workflows:** Users can view found items and request a claim. The claim is locked in a 'pending' state until an admin verifies ownership details and updates it to 'approved' or 'rejected'.
- **Transactional Email Alerts:** Triggered password reset links and verification codes are sent using secure, firewall-bypass HTTP REST API connections (Brevo/Resend) rather than standard blocked port 587 SMTP sockets.

4. Operational Scope & Future Roadmap

The initial release (1.0) concentrates on core MongoDB cloud migration, OAuth mapping, and transactional mail services. Future updates (1.5) will incorporate real-time notifications via WebSockets, Google Maps campus zoning coordinates, and push alerts to mobile web applications.